

A Short Report on Simulation-Based Inference

Scientific Machine Learning Seminar

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Abstract

Simulation-Based Inference (SBI) offers a novel machine-learning-based approach to solve inverse problems. Hereby, a neural network learns to map observed data to a posterior distribution over model parameters. By using simulations, SBI does not require any likelihood evaluation which otherwise often is computationally costly or has to be approximated. This can allow SBI to outperform classical Bayesian inference methods in inference speed and posterior accuracy. However, common problems like model misspecification, neural inaccuracies or miscalibrated posteriors can limit the utility of SBI if they are not circumvented. This report presents basic SBI methodology, in particular Neural Posterior Estimation, discusses its advantages, limitations and reviews two exemplary applications for SBI: detecting planetary events from gravitational microlensing and inferring black hole merger parameters from gravitational waves.

1 Introduction

In computational physics there are usually two types of problems to solve: forward problems and inverse problems. In a forward problem, the parameters and/or boundary conditions of a model are given and the solution is to find data or a data distribution which adheres to this model. For example, in the case of given parameters and initial values, this is a typical simulation.

In contrast, an inverse problem starts from data as a preset and the parameters or parameter distributions of the assumed model have to be inferred. Here, the data likely comes from an observation and the parameters which represent physical properties of the system have to be computed. This is the type of problem that will be explored in this report.

A rather simple case of solving an inverse problem is the use of linear regression to fit a function to a given dataset. For more complex models, however, more efficient methods are needed. The classical approach here is to use Bayesian methods to infer the parameter posterior distribution, but Simulation-Based Inference (SBI) introduces machine learning techniques for solving inverse problems. In specific contexts, SBI is able to solve these problems more efficiently and computationally cheaper. Applications where SBI shows advantages include, but are not limited to, gravitational microlensing (Zhang et al., 2021), gravitational wave physics (Dax et al., 2023) and neuroscience (Deistler et al., 2025, Sec. 4.3).

In Section 2 the general concepts of classical Bayesian inference are first introduced, specifically focussing on Markov-Chain-Monte-Carlo methods. Then SBI and its underlying concepts are introduced in Section 3 and illustrated in Section 4 with the help of a toy-example. The advan-

tages and limitations of SBI are presented in Section 5. Two example applications in astrophysics and their results are presented in Section 6 and a conclusion is given in Section 7.

2 Classical Bayesian Inference

The most outstanding feature of Bayesian Inference (BI), in comparison to regression, is that the output of BI is a posterior distribution of the parameters, rather than simply a likely parameter set from this distribution. Instead of just computing $\theta \sim p(\theta|x)$, the result is the full $p(\theta|x)$ distribution. Here, θ denotes the set of parameters from the model while x is the observed data.

The underlying concept is the Bayes' theorem

$$p(\theta|x) = \frac{p(x|\theta)}{p(x)}p(\theta) \quad (1)$$

from Bayesian statistics where $p(\theta|x)$ is the posterior distribution, $p(x|\theta)$ the likelihood, $p(x)$ the evidence and $p(\theta)$ the prior distribution. While the prior can be chosen adequately according to assumed distributions of the model parameters and the evidence is just a normalizing factor, the likelihood $p(x|\theta)$ has to be calculated, evaluated or approximated when using numerical methods for posterior inference. This is the most important and also most costly step of classical Bayesian Inference (Deistler et al., 2025, Sec. 1).

Different methods of Bayesian Inference approach this differently. Markov chain Monte Carlo (MCMC) methods are relatively simple and widely used for inferring posteriors. These evaluate the likelihood many times at random points in the parameters space and build a posterior distribution from that. In Metropolis-Hastings, a MCMC method, the prior and likelihood value at a random point in the parameter space and their relation to the previous point determine whether this point is accepted or rejected. The accumulated distribution of the accepted points then converges to the posterior after many evaluations (Metropolis et al., 1953; Hastings, 1970).

There are adjusted methods that can make the algorithm more efficient (such as Hamiltonian Monte Carlo or nested sampling) but all classical Bayesian inference methods share similar key limitations. The convergence to a posterior that is accurate enough for the specific application takes many evaluations of the likelihood and is therefore very computationally expensive. For the applications that are later shown in Section 6 this can be up to one second per evaluation, for an adequately sampled posterior distribution the computation time per analyzed dataset can be huge.

In complex models the exact evaluation of the likelihood is often impossible or considerably more computationally expensive. It then has to be approximated by a Gaussian distribution, usually as part of the χ^2 method. This introduces an additional error source into the posterior.

Classical Bayesian inference methods also have the problem that they cannot model multimodal posteriors very well. The algorithm often gets stuck in one mode and is unable to jump to another, this becomes a problem in models with a lot of close degeneracies (Deistler et al., 2025, Sec. 1, 2.1). Consequently, grid searches and human intervention are often needed which increases the time and effort of the data analysis process. This is often impractical for the large amount of data that has to be processed in modern science.

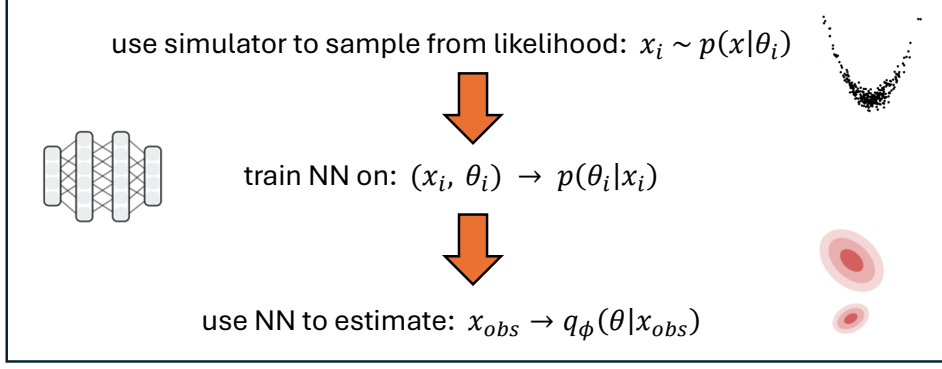


Figure 1: Schematic overview of the SBI-workflow. Three steps are shown: simulation of a dataset (θ_i, x_i) , training of the neural network (NN) to estimate the true posterior density $p(\theta_i, x_i)$ and posterior estimation which maps the observed data x_{obs} to an estimated posterior distribution $q_\phi(\theta|x_{\text{obs}})$. Parts of the figure are adapted from [Deistler et al. \(2025\)](#).

3 Introduction to Simulation-Based Inference

The key feature of Simulation-Based Inference is that it avoids the computationally expensive likelihood evaluation. This is the reason SBI is sometimes also called Likelihood-Free Inference. Instead of explicitly evaluating the likelihood at specific points in parameter space, the likelihood is implicitly encoded in the data sampled from a simulator of the model. A neural network is trained on this simulated dataset to infer a parameter posterior distribution from an observation. The likelihood still exists inside the dataset but does not have to be calculated or approximated.

This report focusses on the direct inference of posteriors, a methodology called Neural Posterior Estimation (NPE). In contrast, Neural Likelihood Estimation (NLE) and Neural (Likelihood-)Ratio Estimation (NRE) each require an additional step of computation to obtain the posterior but may offer further advantages or more insight into the analysis process.

The SBI workflow can be subdivided into three steps as shown in Fig. 1: First, the simulator has to generate a relatively large dataset in accordance with the physical model. To achieve this, an adequate parameter prior distribution is chosen based on assumptions about the physical properties of the model, e.g. the range of specific parameters. The simulator then samples x_i from the likelihood $p(x|\theta_i)$ thereby generating a (x_i, θ_i) dataset. The likelihood is implicitly contained in the dataset, but because the simulation solves a forward problem, this step is not very computationally costly.

Next, a neural network is trained to map each tuple (x_i, θ_i) to an estimated posterior density $q_\phi(\theta_i|x_i)$. The network can in principle be chosen freely, but commonly an architecture based on Normalizing Flows is used. The loss function

$$\mathcal{L}(\phi) = \mathbb{E}_{(\theta_i, x_i) \sim p(\theta, x)} [-\log q_\phi(\theta_i|x_i)] \quad (2)$$

is minimized to optimize the network weights. It is the expectation value, over all sampled tuples, of the negative logarithm of the estimated posterior density. This closely resembles the definition of a cross entropy and balances low global loss values against high confidence in the posterior during training. Since $q_\phi(\theta_i|x_i)$ is normalized, maximizing it at the correct θ_i has the drawback of having to be more confident by narrowing the distribution. Thus, the estimated posterior density $q_\phi(\theta_i|x_i)$ is trained towards representing the true posterior $p(\theta_i|x_i)$.

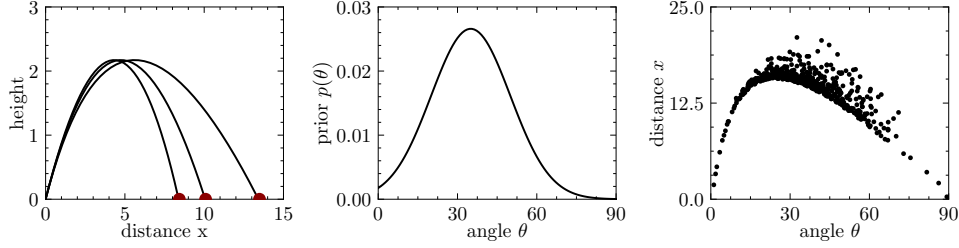


Figure 2: Simulations in SBI from the example of an angled throw. Left: simulation of throwing trajectories, Center: prior distribution of parameters θ_i , Right: simulated dataset (θ_i, x_i) . The example is adapted from [Deistler et al. \(2025\)](#).

Lastly, the observed data point x_{obs} is fed into the network, which is evaluated over an interval of θ at any resolution. This yields the posterior distribution of any observation at a relatively low computational cost. It is worth noting that there is no need for further training when analyzing new data under the assumption of the same model ([Deistler et al., 2025](#), Sec. 2.2, 3).

4 Simple SBI Example

A fairly simple demonstration of SBI is done in this section with the help of the simulation of an angled throw. The structure and code of this toy-example are adapted from [Deistler et al. \(2025\)](#).

A ball is thrown from the origin at a specific angle θ . The throwing velocity is always the same so θ is the sole parameter of the model. The ball lands on the ground at a specific distance from the origin, again this is the only data value x . The system behaves according to the physical laws of gravity, the flight trajectory is a parabola. To turn the model probabilistic and make it harder for SBI to infer the posterior, a varying wind speed is added as a noise source.

Now it is not trivial to infer the posterior distribution of throwing angles from just a single distance measurement. Even classical Bayesian methods would have problems, since the posterior is bimodal: a short throw could be the result of either a very low angle or very high angle. A few example trajectories are shown in [Fig. 2](#) (left).

A parameter prior for throwing angle θ also has to be chosen. Here a Gaussian distribution truncated to $[0^\circ, 90^\circ]$ with its maximum at 35° (as shown in [Fig. 2](#), center) is a reasonable choice. Parameter values θ_i are sampled from this prior and put into the simulator that contains the probabilistic model of the angled throw as described above. Many angle–distance tuples (θ_i, x_i) are generated, building the dataset that is shown in [Fig. 2](#) (right).

This dataset is used to train a neural network based on the architecture of Normalizing Flows. The network takes a tuple (θ_i, x_i) as input and outputs the predicted posterior density value q_ϕ at specifically $\theta = \theta_i$ in the parameter space under the assumption $x = x_i$. This is shown as a schematic in [Fig. 3](#). The network is optimized using the loss function in [Eq. \(2\)](#) as explained in [Section 3](#).

With the trained network a posterior distribution is obtained by supplying the observation x_{obs} , e.g. $x_{\text{obs}} = 13\text{ m}$, and evaluating the posterior density over an interval of angles θ . This is repeated for a few hypothetical observations and sampled for $0^\circ < \theta < 90^\circ$. The sampling rate of θ can be chosen freely; it increases quality and affects the computational cost, but a finer grid brings no benefit beyond the accuracy of the network’s estimate. In this simple example,

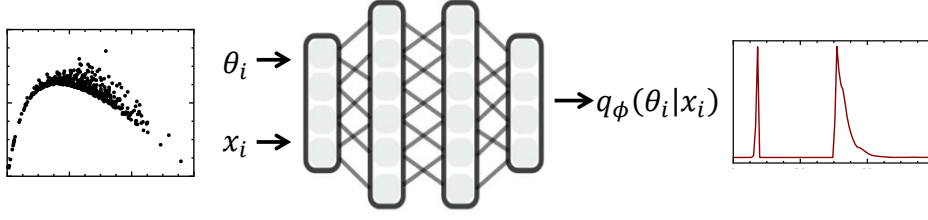


Figure 3: Training of the neural network in SBI from the example of an angled throw. The network learns to map the dataset of tuples (θ_i, x_i) to the estimated posterior density $q_\phi(\theta_i|x_i)$. The example and parts of the figure are adapted from [Deistler et al. \(2025\)](#).

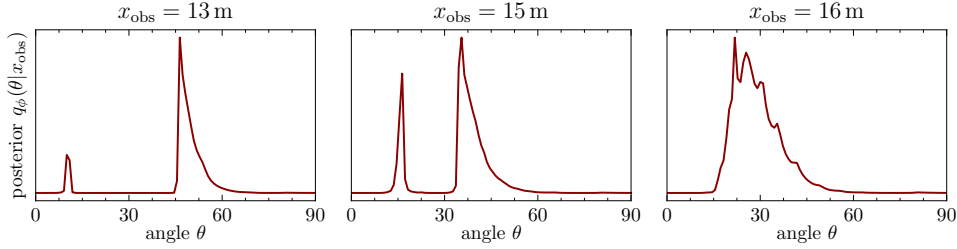


Figure 4: Estimated posteriors distributions $q_\phi(\theta|x_{\text{obs}})$ achieved with SBI from the example of an angled throw. The trained neural network was sampled at the specified x_{obs} values and evaluated in a θ -intervall between 0° and 90° . The example is adapted from [Deistler et al. \(2025\)](#).

however, the cost is negligible.

Results are shown in Fig. 4. For smaller distances x_{obs} , the two modes in the posterior are easily visible as two separate peaks, one with low angles and one with high angles. For a high x_{obs} value of 16 m the estimated posterior is noisy, a consequence of a limited number of simulations and subsequently a low density of datapoints at this distance.

5 Benefits and Limitations

SBI is able to outperform classical Bayesian Inference in specific contexts. The method’s benefits can be summarized into three main concepts, which are presented in this section, followed by common limitations of SBI.

First, the inference of the posterior is likelihood-free. There is no need to explicitly evaluate the likelihood at random points in the parameter space like it is the case for MCMC, the likelihood is implicitly contained inside the dataset from the simulator the neural network is trained on. Inference itself can therefore be made much faster, and no likelihood approximations, as often needed for Bayesian Inference, are required.

Second, SBI enables Amortized Inference. Here, most of the computational cost is paid upfront, when running the simulator and training the neural network, while the actual inference by the network is very fast and cheap. This can greatly accelerate the analysis when one model is reused for a lot of observations, since no further simulation and training is needed once the network weights are optimized. For complex models this is especially relevant, classical Bayesian Inference would require a considerable amount of computation time at each likelihood evaluation.

Third, SBI is able to capture the multimodality of posteriors very well. Unlike in MCMC, where

the sampler can get stuck in a single mode and modes often have to be explored manually, the network represents the entire posterior at once rather than relying on local likelihood evaluations.

Though bundled with these advantages, there are also limitations come with SBI algorithms. The previously mentioned upfront simulation and training due to Amortized Inference may be a drawback when there is only a limited amount of observations to be processed. Therefore, the gain in inference speed may not justify the upfront cost. Additionally, a change in the parameter prior requires an entirely new costly simulation and training run.

Since SBI is only as good as the simulator, a common pitfall is model misspecification. When the neural network is given an observation that has not been simulated before, because such a case is not included in the model, the output will be a nonsensical posterior distribution that is not physically meaningful. To illustrate with the example in Section 4, for a distance larger than any possible throw, the network will still return an unpredictable non-zero posterior over angles, even though no angle could lead to such a distance. In complex models, this is not trivial to detect and avoid.

It is important to remember that the output posterior is only a neural estimation and its quality depends on the simulations, the network architecture as well as training. There is no convergence guarantee analogous to MCMC to assure that the estimated posterior is getting closer to the true posterior. MCMC becomes asymptotically exact for long chains, whereas the NPE estimate is exact only in the limit of infinite training data and network capacity, which makes its quality difficult to assess.

A common problem of posteriors inferred via NPE is miscalibration. The posteriors can be biased (a shifted mean to either side), overconfident (too narrow) or underconfident (too broad). However, miscalibration can be diagnosed effectively with quantile-occurrence plots. These compare the posterior probability assigned to a tuple with how often it actually occurs in a validation dataset, thereby revealing errors in posterior accuracy on a global scale. The quantile-occurrence plots for model parameters from an application by Zhang et al. (2021) presented in Section 6 are shown in Fig. 5 (right). Both bias and underconfidence are present in these plots, noticeable by their characteristic histogram shapes. If the plots indicate miscalibration, changes to the network and training are probably necessary (Deistler et al., 2025, Sec. 1, 2.2, 3).

6 Application

The research area of gravitational microlensing is a prime example where SBI shines in the data analysis pipeline. When a star with an exoplanet in orbit moves into the line of sight between Earth and a bright background star, the light of background star is lensed around the foreground star and exoplanet and magnified in a characteristic way. This effect cannot be spatially resolved, but telescopes can measure the change of brightness as a light curve. The shape of the light curve is defined by the physical properties of both stars and exoplanet, resulting in a large amount of parameters (around 11) and a complex model, while the data is often limited. Additionally, the model is highly degenerate and often has multiple complex modes that have to be explored manually when using classical Bayesian Inference.

This is the reason why Zhang et al. (2021) tested the capabilities of SBI on microlensing data from the Roman Space Telescope. The training was done on a dataset of more than 275,000 simulated events, which took about 25 hours. SBI was able to increase the computation speed drastically, from 1 simulation per second, which would be needed for one MCMC step, to 10^5

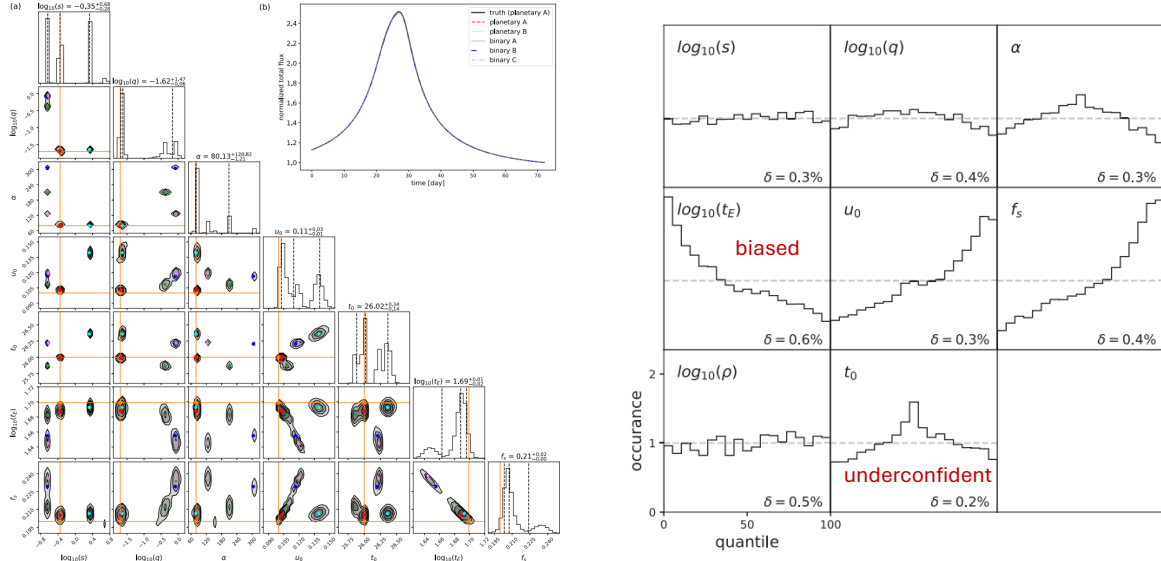


Figure 5: Results from SBI application on gravitational microlensing by Zhang et al. (2021). Left: Corner plot of posterior distributions obtained with SBI (a) and simulated lightcurves (b). The posterior has five modes which are modeled accurately with the true value (orange) close by one of these. Right: Quantile-occurrence plots for parameters used in the model. $\log_{10}(t)$, u_0 and f_s are biased, while t_0 and α are underconfident. The figures are taken from Zhang et al. (2021).

posterior samples per second. All posterior modes were modeled accurately, although some biases existed in the posterior. A corner plot of an exemplary posterior (left) as well as quantile-occurrence plots (right) are shown in Fig. 5. Noticeably, the five modes of that posterior were captured by SBI and are easily discernable, but clear biases exist in the posteriors of $\log_{10}(t)$, u_0 and f_s , while t_0 and α show underconfidence. The authors recommend following up on SBI with MCMC to refine the posterior, get rid of biases and identify the most likely solution. This demonstrates the usefulness of SBI in the data analysis pipeline for quickly exploring multimodality of posteriors and obtaining initial values for further inference methods.

Similarly, the models needed to analyze gravitational wave signals from black hole mergers depend on a lot of parameters, while only very limited data is available. Dax et al. (2023) also use SBI in their data analysis pipeline for gravitational wave events, but implement Flow Matching Posterior Estimation (FMPE) in their SBI architecture. Instead of Normalizing Flows, FMPE is based on a Flow Matching neural network. It learns a vector field transformation that turns a simple distribution to the real posterior, rather than transforming the data points. This requires fewer parameters to describe the same transformation, yields a more expressive neural network and reduces training time by 30%. Conversely, the inference time is increased, since the sampling also requires solving an ordinary differential equation defined by the velocity field output.

Generally, all data-intensive fields of physics or other science research would likely benefit from an implementation SBI in their data analysis pipelines. Other fields noted in Deistler et al. (2025) for SBI application are psychophysics and neuroscience.

7 Conclusion

Simulation-Based Inference offers benefits that are advantageous in specific parts of data analysis pipelines across many areas of physics. Namely, these are the Likelihood-Freeness of the Method,

Amortized Inference and the ability to capture multimodality in posteriors accurately. Limitations include the possibility of misspecification in the simulation, miscalibration of the posterior inside the neural network and the neurally approximated nature of the output posterior. These must be monitored and mitigated with validation techniques such as quantile-occurrence plots.

An ideal use case for SBI would be an inverse problem where the inference is time-critical, a likelihood evaluation is expensive and the posterior is possibly multimodal. Within a data analysis pipeline, SBI is valuable for obtaining a globally accurate estimate of the posterior. It can be used to explore the parameter space and can be refined by expensive classical Bayesian methods at specific, very likely solutions, like it was recommended by Zhang et al. (2021).

As datasets in physics and other sciences expand rapidly, faster yet still accurate ways of inferring information are needed, and Simulation-Based Inference is going to be a very useful machine-learning-based methodology in that regard. It is reasonable to expect that the use of SBI in physics will increase in the coming years. Open challenges remain and improvements are still to be made, but SBI will presumably grow into a key part of the modern Bayesian data analysis toolkit.

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